The Humanitarian Planner's Dilemma

Rapid, Evidence-Grounded Crisis Response Under Data Scarcity

Aaron Scherf • Columbia University | Sustainable Development (SIPA) • *3R Framework: Resilience, Response, and Recovery*

The Humanitarian Planning Problem

Dual Realities Confronted Simultaneously in Crisis Response

⏱️ Operational Reality (The Planner)

Response Window: 2 – 3 Weeks

- **Immediate Resource Allocation**
  Must deploy aid packages before famine peaks.
- **Hard Budget Constraints**
  Fixed funds; must pick optimal intervention mix.
- **Zero Fieldwork Feasibility**
  Cannot run 6-month survey or pilot RCT in war.
- **Current Fallback**
  Subjective heuristics & qualitative tabletop guessing.

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📚 Evidence Friction (The Literature)

10,000+ Available Evaluations

- **Prohibitive Search Costs**
  Buried across 3ie, J-PAL, World Bank, and paywalls.
- **Complex Econometric Jargon**
  Written for methodologists, not field directors.
- **Single-Program Isolation**
  Isolates one program; ignores opportunity costs.
- **Uncertain Local Relevance**
  Does peacetime cash evidence hold in blockaded crises?

The Reality of Crisis Response: Search Costs & Portfolio Decisions

- **Evidence Search & Consumption Costs:**
  - Thousands of evaluations exist (3ie, J-PAL, World Bank), but they are locked in fragmented repositories and 80-page PDFs.
  - Responders cannot consume, evaluate, or standardize them during an unfolding emergency.
- **The Portfolio Optimization Problem:**
  - Academic papers isolate single interventions for clean internal validity.
  - Responders face an **open canvas and fixed budget**—needing to determine the *optimal mix* of cash, food aid, vouchers, insurance, and logistics.
- **The Resilience Planning Parallel:**
  - Planners face severe data scarcity (no live field surveys).
  - Almost always planning for **unprecedented local shocks**—inherently requiring evidence borrowed and adapted from other contexts.

Divergence in Mandate: Planner vs. Researcher

Real-World Portfolio Optimization vs. Internal Causal Identification

The Humanitarian Planner Mandate

Core Objective:
Maximize lives saved under hard budget caps.

Primary Question:

"Given $5M and 3 weeks in District Z, what mix of A, B, or C minimizes loss?"

Intervention Framing:
Open portfolio choice & program layering.

Time Horizon:
Days or weeks before emergency peak.

Tolerance for Uncertainty:
Satisficing: Needs honest probabilistic bounds.

✓ Goal: Simulation Response Exercise

The Academic Research Mandate

Core Objective:
Isolate single causal parameter (internal validity).

Primary Question:

"Does Intervention X cause Outcome Y?"
(Holding all other contextual factors constant)

Intervention Framing:
Single intervention evaluated in isolation.

Time Horizon:
3 to 5 years (design, survey, peer review).

Tolerance for Uncertainty:
Low: Prefers no estimate over an unverified proxy.

✓ Gold Standard: Clustered RCT

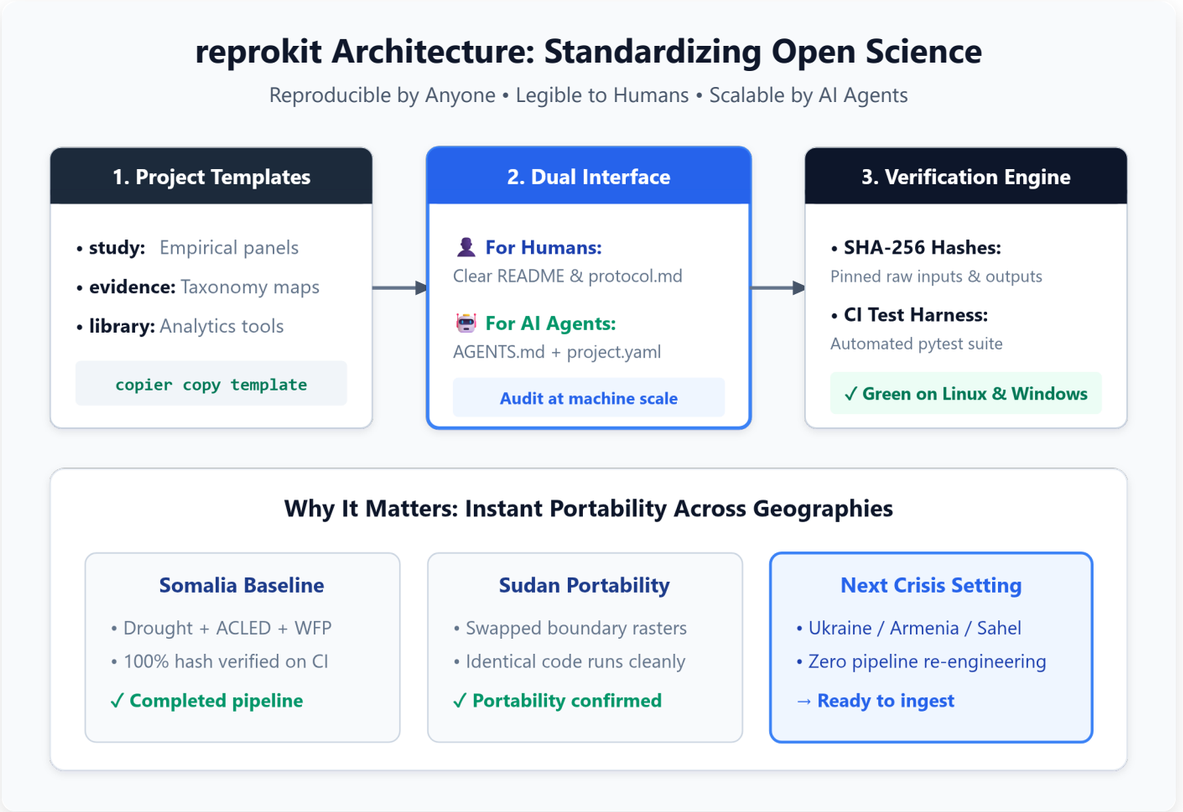
State of the Art: The Limitations of Existing Frameworks

- **3ie Evidence & Gap Maps:**
  - Excellent taxonomies, but static and manually curated.
  - Disconnected from live spatial data; does not extract reusable estimators.
- **World Bank DIME / Microdata & HDX:**
  - Rich passive data repositories lacking counterfactual simulation engines.
- **Academic Hierarchical Modeling:**
  - Meager (2019) & Vivalt (2020): Prove high cross-study effect variance across microcredit and social programs.
  - Pritchett & Sandefur (2015): External trial estimates often yield higher error than local non-experimental data.
  - *Limitation:* Retrospective academic studies on clean survey microdata; inapplicable to rapid, live crisis response.



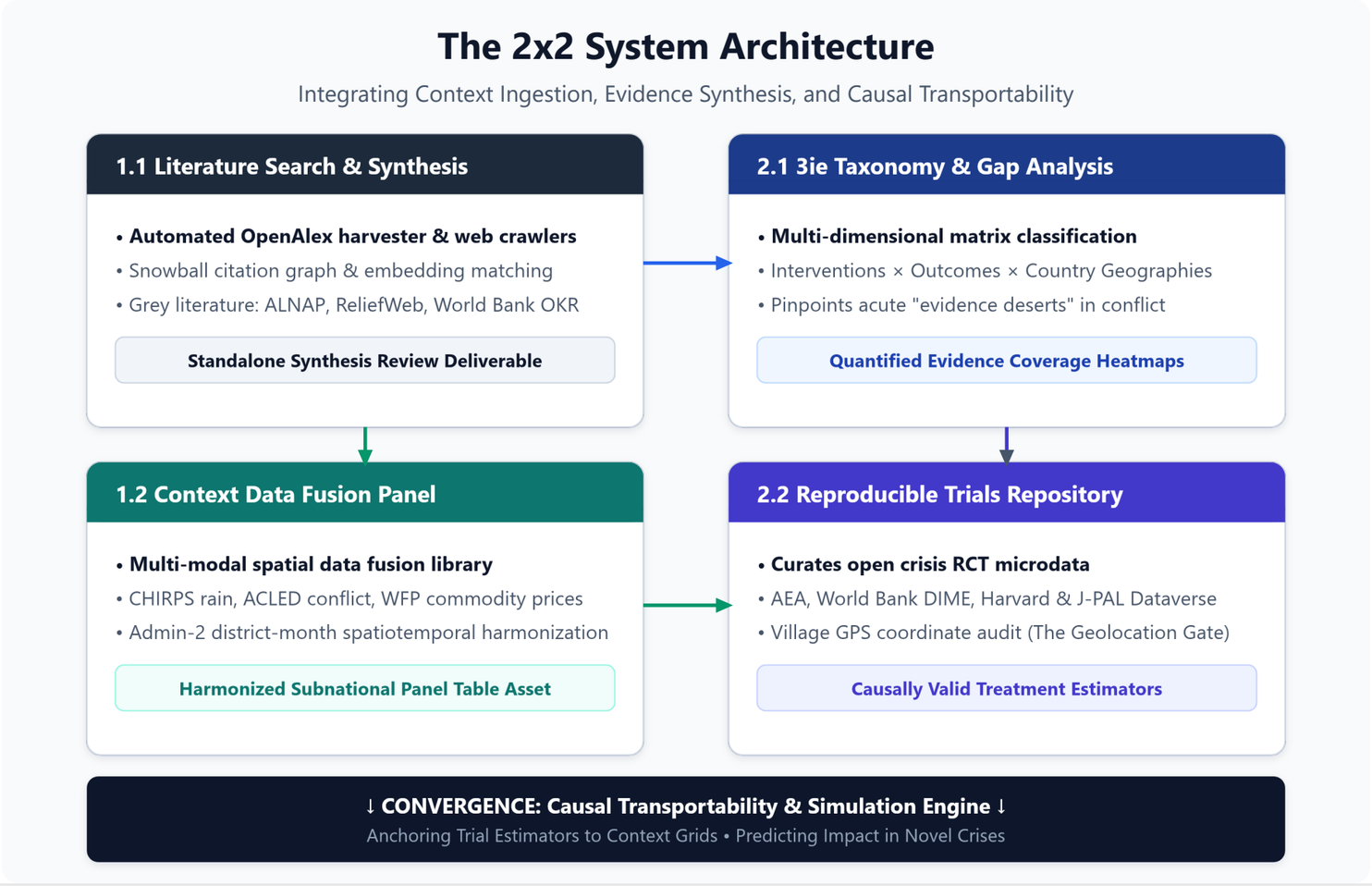
Foundational Architecture: Standardized, Reproducible Research (`reprokit`)

- **Standardizing Open Science & Data Science:**
  - Orchestrated via `reprokit` (v0.4.0 released, open-source).
  - Independent verification regardless of technical background.
- **Three Project Archetypes:**
  - `study`: Empirical pipelines with automated hash verification.
  - `evidence`: Machine-readable evidence maps and manifests.
  - `library`: Reusable data fusion and analytics packages.
- **Built for Human and Agent Readability:**
  - Standardized metadata (`project.yaml`, `AGENTS.md`) allows autonomous LLM agents to parse and audit codebases at scale.
  - Dual-interface: accessible to humans, verifiable by agents.
- **Portability Across Contexts:** Hash-pinned inputs and CI on Linux/Windows enable rapid porting to new crises.



The 2x2 System Architecture

Integrating Context Ingestion, Evidence Synthesis, and Causal Transportability



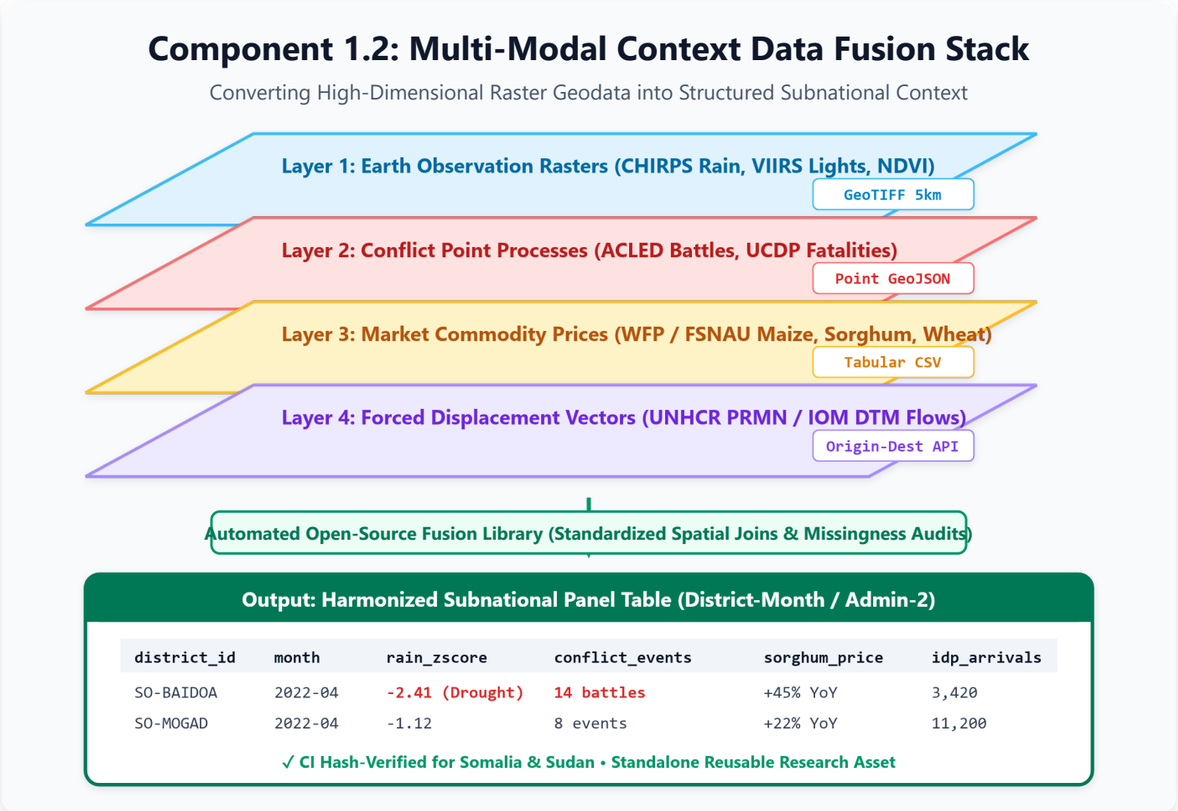
Component 1.1: Automated Literature Discovery & Synthesis

- **Overcoming Search Costs:**
  - Automated scrapers continuously harvest OpenAlex, academic APIs, and grey literature portals (ALNAP, ReliefWeb, World Bank OKR).
- **Snowball & Semantic Graph Expansion:**
  - Traces forward/backward citation networks around key crisis topics.
  - Extracts metadata, abstract embeddings, and author linkages.
- **Standalone Usable Deliverables:**
  - Not just an intermediate step: produces a **comprehensive, publishable literature review** and evidence synthesis on demand for any designated geography.
- **Current Status:**
  - ACTIVE Automated harvesting running; expanding from ad-hoc sampling to systematic corpus extraction.



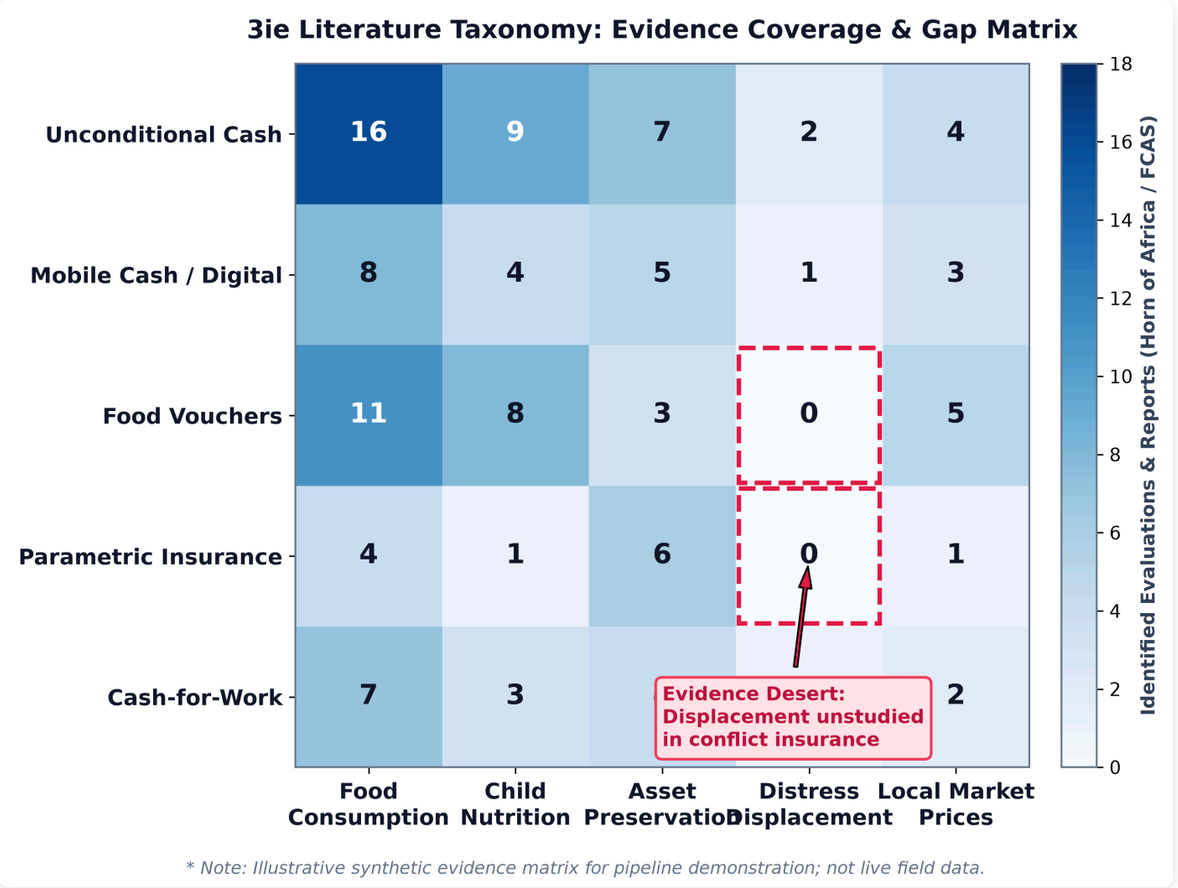
Component 1.2: Subnational Context Data Fusion & Analytics Library

- **Standalone Research Deliverables:**
  - The harmonized panel dataset and the processing pipeline are valuable standalone research assets.
- **The Open-Source Data Fusion Library:**
  - Automates multi-modal ingestion (satellite rasters, point events, price feeds).
  - Standardizes spatiotemporal boundaries (Admin-2 / district-month).
  - Automated cleaning, spatial joins, and missingness audits.
- **Core Technical Contribution:**
  - Transforms high-dimensional Earth observation rasters (CHIRPS precipitation, night lights) into structured, tabular context variables alongside conflict logs (ACLED/UCDP) and market feeds (WFP).
- **Proof of Concept:** Somalia and Sudan pipelines rebuilt with 100% hash matching on CI.



Component 2.1: Quantifying Evidence Coverage via 3ie Taxonomies

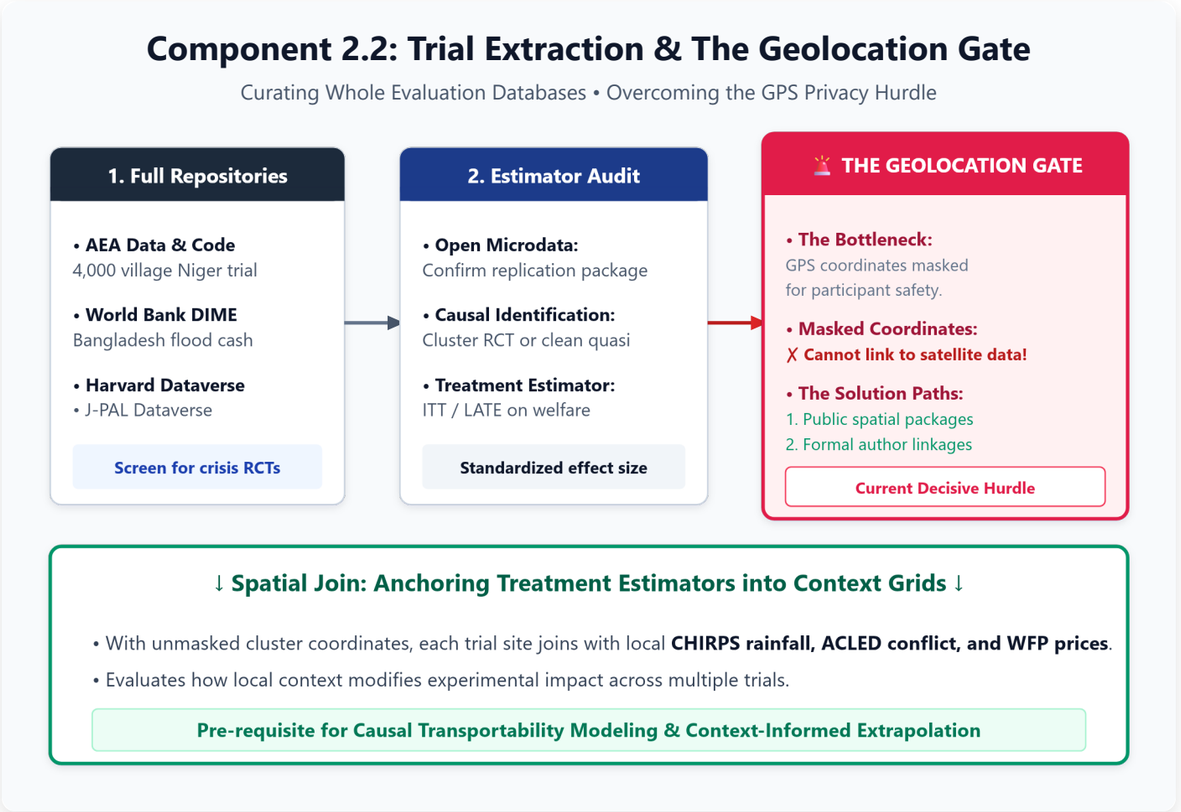
- **Multi-Dimensional Evidence Matrices:**
  - Categorizes academic and institutional grey literature using 3ie's standardized intervention (cash, vouchers, insurance, public works) and outcome taxonomies (consumption, nutrition, displacement).
- **Country-Specific Granularity:**
  - Generates distinct evidence heatmaps for any target country of interest (starting with Somalia and Sudan testbeds).
- **Diagnosing "Evidence Deserts":**
  - Dense green zones: Cash in peaceful agricultural settings.
  - Blank/red deserts: Parametric insurance, anticipatory cash, or social safety nets under active conflict.
- **Current Status:**
  - ACTIVE Proof-of-concept classification complete on initial paper sample; scaling across the corpus.



** Note: Illustrative synthetic evidence matrix for pipeline demonstration; not live field data.*

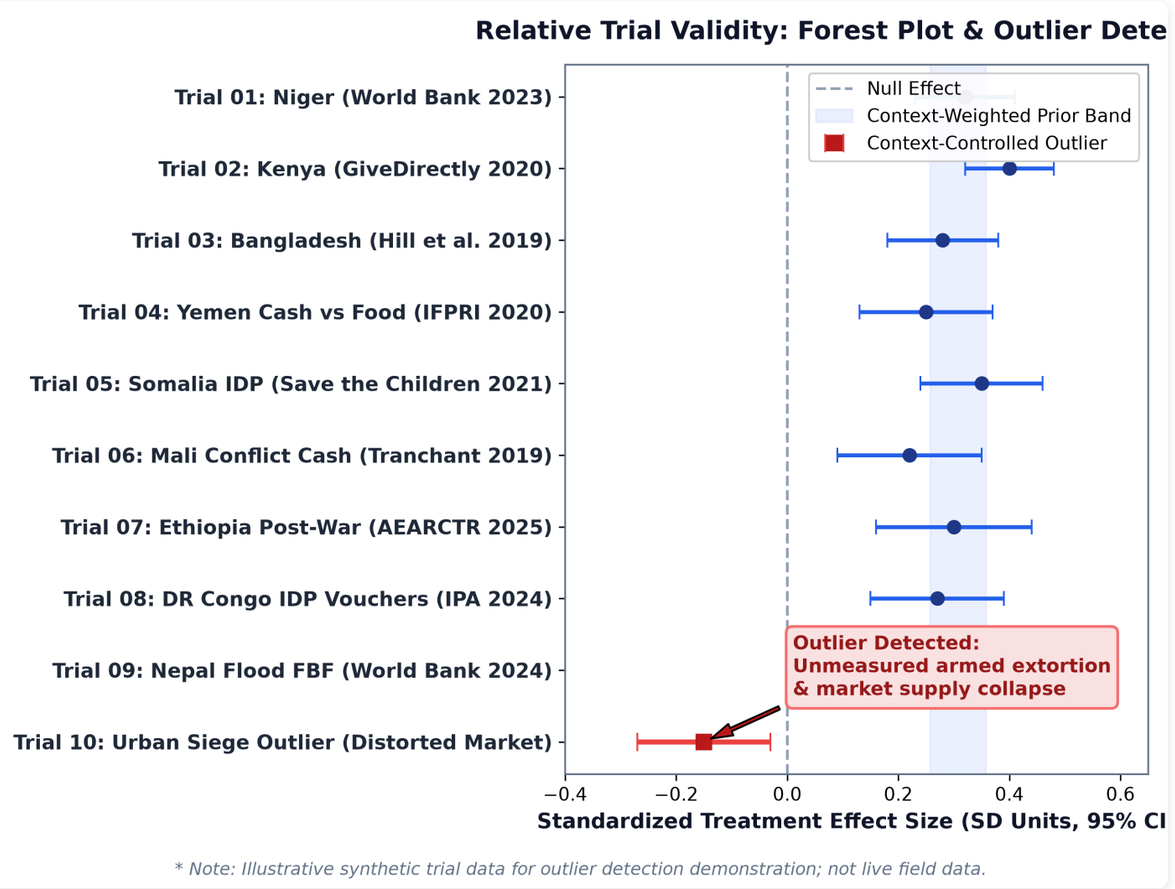
Component 2.2: Curating Whole Trial Repositories & The Geolocation Gate

- **Candidate Corpus: Whole Evaluation Databases:**
  - Curating trial data from complete public databases: **AEA Data and Code Repository**, **World Bank DIME / Microdata Catalog**, and **Harvard / J-PAL Dataverse**.
  - Initial test cases: Cash transfer evaluations in Niger (4,000 villages), Bangladesh flood response, and Horn of Africa safety nets.
- **The Geolocation Bottleneck (The Gate):**
  - **The Challenge:** To link an experimental trial with local satellite and conflict context, we require village or cluster GPS coordinates.
  - **The Hurdle:** Many public trial datasets mask GPS coordinates to protect participant privacy in fragile contexts.
- **Resolution Strategy:**
  - 1. Public spatial replication packages.
  - 2. Author outreach and institutional agreements.



Convergence & Empirical Validation: Relative Trial Validity

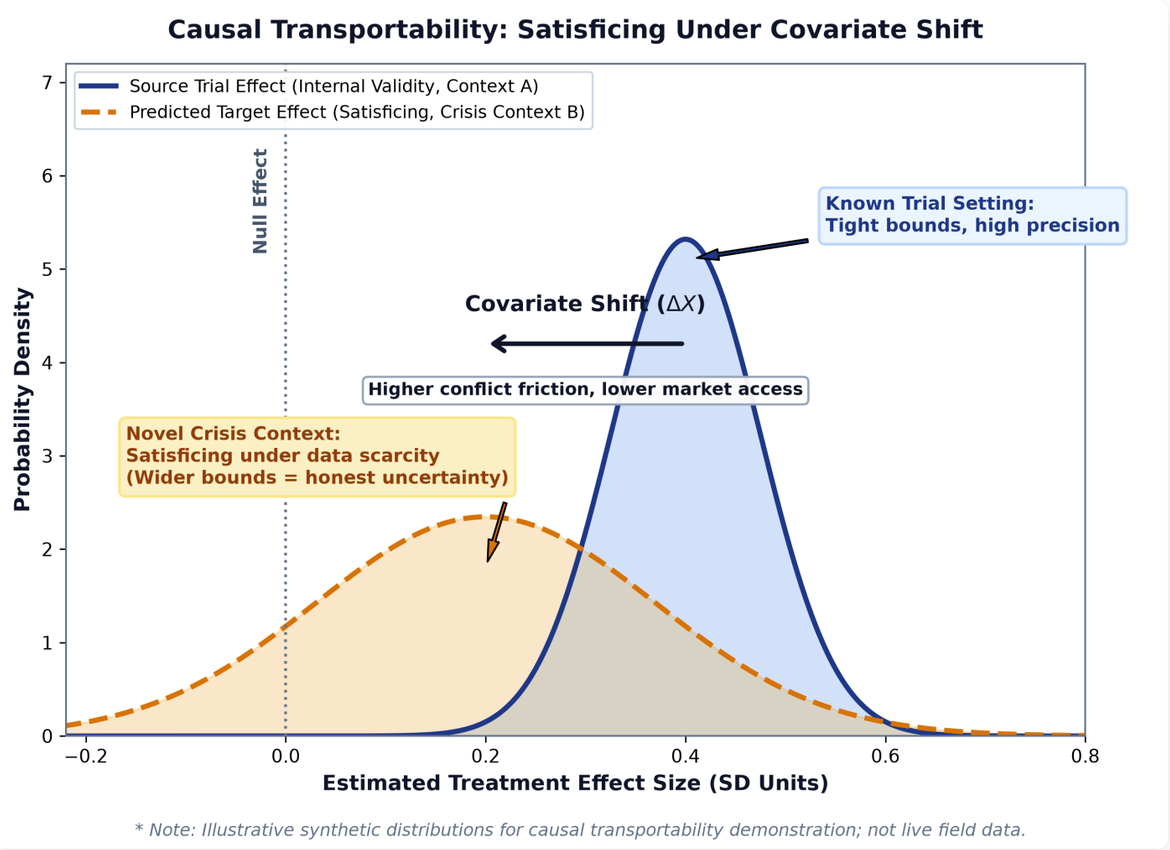
- **The Feedback Loop Between 1.2 and 2.2:**
  - Unmasked trial sites merge with our subnational context panel (rainfall anomalies, market prices, local conflict intensity).
- **Relative Trial Validity Check:**
  - Instead of assessing a trial in a vacuum, we evaluate its treatment effect relative to similar trials across different contexts.
- **Automated Outlier Detection:**
  - *Example:* If an urban siege trial shows negative welfare effects while 9 rural cash trials show positive effects, the model flags an outlier.
  - *Interpretation:* Unmeasured market collapse or extortion distorted the expected causal mechanism.
- **Refining the Prior:** Outliers feed back into the 2×2 engine, flagging unmodeled structural variables.



** Note: Illustrative synthetic trial data for outlier detection demonstration; not live field data.*

Causal Transportability Under Covariate Shift: Satisficing in Crises

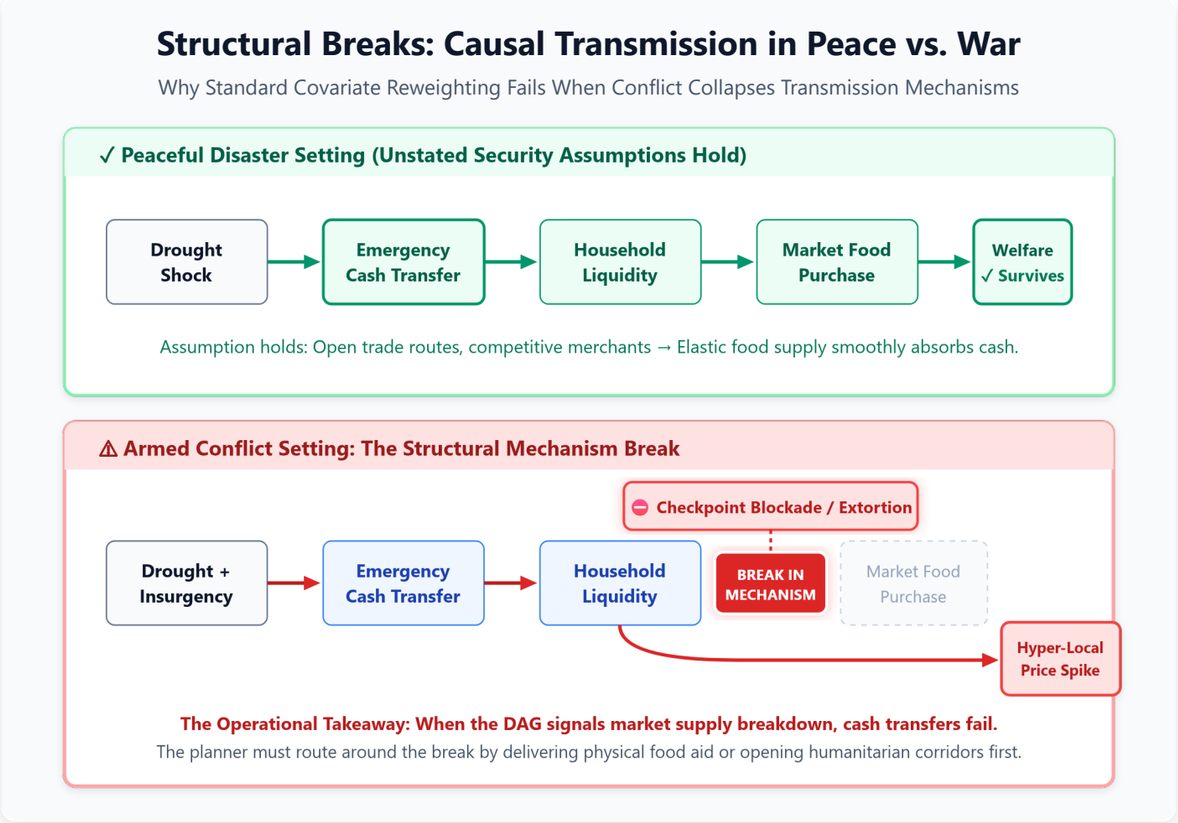
- **The Baseline Challenge: Covariate Shift:**
  - External transportability fails even in peaceful settings: populations, market density, and baseline wealth differ across borders.
  - Interventions are held constant, but the background environment changes ($X_{source} \neq X_{target}$).
- **Satisficing Under Data Scarcity:**
  - In active conflict and disasters, perfect identification is impossible because field surveys cannot be conducted.
  - **The Goal is Satisficing:** Generating the best possible empirical estimates of potential impact with valid, honest confidence intervals.
- **Quantifying What Does NOT Apply:**
  - Measuring the degree to which past evidence does *not* translate, modeling effect variability from known trial dispersions.



** Note: Illustrative synthetic distributions for causal transportability demonstration; not live field data.*

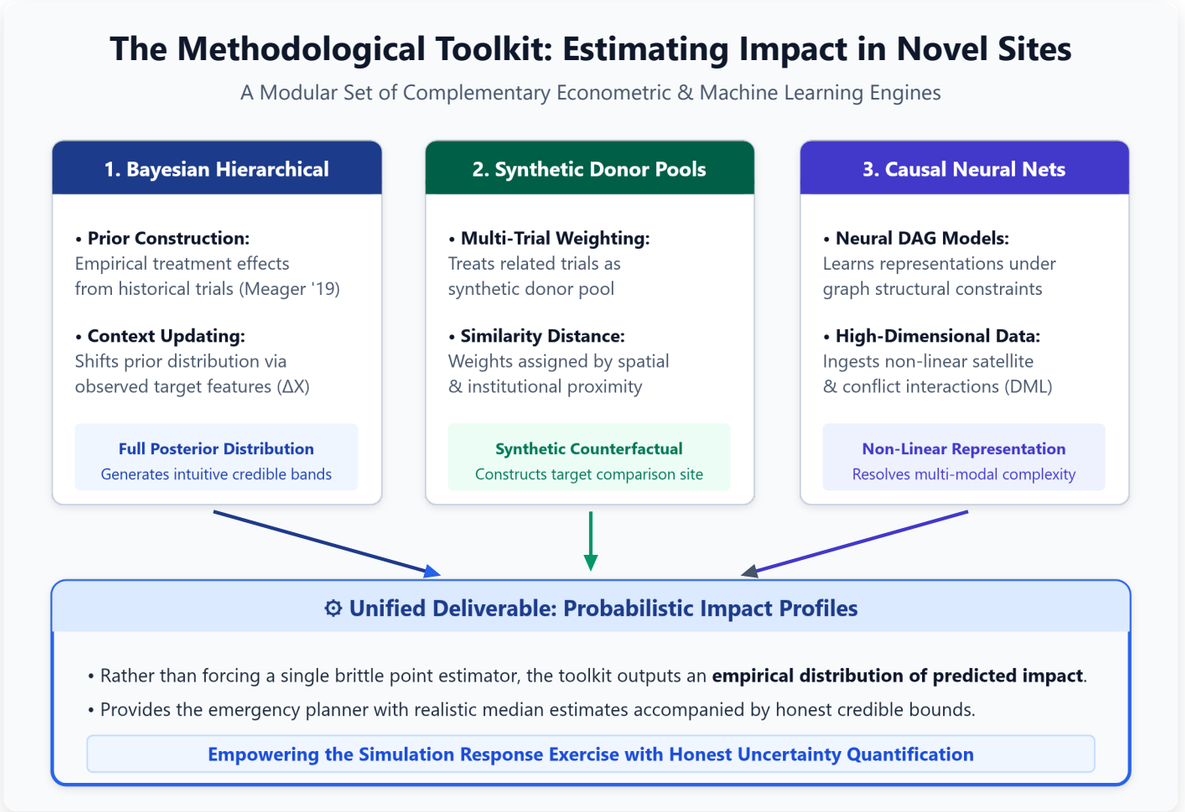
Structural Breaks in Causal Linkages: When Mechanisms Collapse

- **Beyond Covariate Shift: The Structural Break:**
  - *In Peace:* Cash to farmers increases drought resilience, assuming open trade corridors, functional mobile networks, and non-predatory authorities.
  - *In Conflict:* Armed groups extort recipients, mobile networks are jammed, or trade routes are blockaded. Causal linkages collapse.
- **Structural Break as a Validity Check:**
  - An essential audit of whether the causal chain itself is intact before attempting covariate adjustment.
- **Modeling Breakdowns with Directed Acyclic Graphs (DAGs):**
  - Mapping nodes: Transfer → Liquidity → Market Purchase → Welfare.
  - Identifying friction nodes: Military blockades, inflation spikes, and extortion.
- **Operational Route-Arounds:** If the market link is broken (inelastic supply), planners know cash will fail and in-kind food distribution is required.



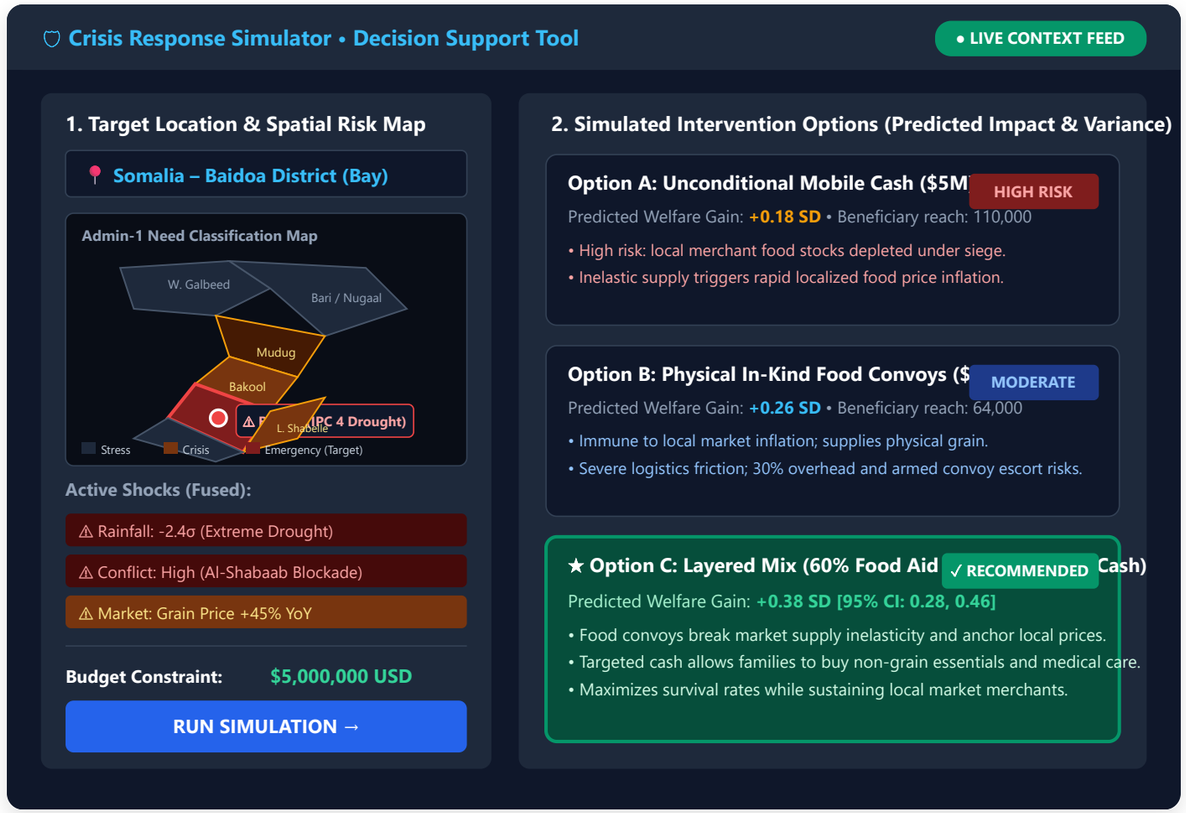
The Methodological Toolkit: Estimating Impact in Novel Localities

- **A Modular Set of Complementary Tools:**
  - Not a single rigid equation, but a flexible toolkit tailored for different data environments.
- **Bayesian Hierarchical Modeling (e.g., Meager 2019):**
  - Priors from past trials, updated by observed context shifts (from fusion panel) to yield posterior distributions for the target site.
- **Synthetic Controls / Multi-Trial Weighting:**
  - Weighting historical trials as a synthetic donor pool based on spatial, climate, and institutional similarity.
- **Double Machine Learning (DML; Chernozhukov et al.):**
  - High-dimensional regularized estimation to control for complex spatiotemporal confounders.
- **Causal Neural Networks & Neural DAGs:**
  - Learning representations under structural graph constraints, modeling non-linear interactions across satellite rasters.



Decision Support in Action: The Simulation Response Exercise

- **The Simulation Response Exercise:**
  - Runs the transport engine across candidate interventions for a target district under hard budget constraints ($5M, 3-week window in Baidoa).
  - Simulates predicted welfare gains, cost-effectiveness, and uncertainty bounds.
- **Overcoming Inelastic Supply Corridors:**
  - *Option A (Pure Cash):* Rapid delivery, but high risk of triggering hyper-local price inflation under armed blockade.
  - *Option B (Food Aid Convoys):* Immune to local grain inflation, but heavy logistical overhead.
  - *Option C (Layered Portfolio - Recommended):* Food convoys break supply inelasticity, allowing cash to restore family livelihoods.
- **Active Real-World Engagement:**
  - Collaborating with a foundation in Armenia to deploy this structured evidence mapping and decision simulation toolkit for emergency planning.



Methodological Frontiers, Open Questions & Roadmap

- **Core Theoretical & Empirical Frontiers:**
  - **Intervention Combining & Layering:** Modeling positive complementarities when cash, food aid, and insurance are combined.
  - **Displacement & Mobility Dynamics:** Does emergency aid facilitate safe evacuation or cause entrapment under armed siege?
  - **Spatial Resolution Limits:** Resolving conflict events and mobility vectors below Admin-2 to match trial clusters.
  - **Proxy Surrogacy in Conflict:** When do satellite indicators (NDVI, night lights) fail as proxies for welfare?
- **Roadmap & Planned Milestones:**
  - **Nov 2026 (Now):** Case study replication, reprokit v0.4.0 live, SusDev Brownbag, and geolocation data audit.
  - **Spring 2027 (Evidence Scaling):** 2x2 matrix synthesis, OpenAlex corpus sweep, SusDev Seminar talk, and Working Paper 1.
  - **Summer 2027 (Transport Models):** Formal causal transportability framework (covariate shift & DAG structural breaks).
  - **Fall 2027 (Applied Case Test):** Simulation response exercise tested against real-world crisis scenario.

